Winkel Board: A Powerful, Integrated Solution For Makers

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While building an Internet of Things (IoT) product, there is one thing that consumes a lot of time. Despite being a vital part, finding the right components to use and figuring out ways to interface these is always a huge task. To get started, it requires knowledge of various chips and libraries, a powerful set of input/output (I/O) pins and a simple integrated development environment (IDE).

Any engineer at some point or the other has yearned for a magic wand that can quickly accelerate the initial design stage so that he or she can quickly get on with actual building or prototyping. This wish was heard by Rishi Hedge, a maker from Pune and founder of Mintbox Technologies, who built a unique Arduino-compatible open source hardware development platform for makers and hardware hackers, called Winkel Board.

Overcoming the usual mess

Arduino has proven its usefulness to most makers and product developers in the recent past. The actual problem arises when a project demands various shields. In all IoT projects, connecting to the Internet is the vital part.

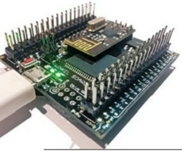
Connecting Arduino to the Internet requires an Ethernet or Wi-Fi shield. You may require accurate timing in your project so that the controller can fulfil the tasks at particular intervals, which means that your controller should be connected to a real-time clock.

If you want to create a mesh network using radio nodes, you need to know how to get started with NRF24l01 or, alternately, a radio shield. Similarly, interfacing Bluetooth requires a separate Bluetooth shield.

In India, shields are expensive, and sourcing these from abroad affects the speed of prototyping to a huge extent. Despite that, you can imagine the number of wires and modules and space used and, of course, the mess created in a project that uses all these shields.

This was, in fact, one problem faced by the makers of Winkel Board, which they soon found out to be a common problem among makers. "Some suggested Raspberry Pi 3 to be the answer to many projects. But not every project needs an operating system and not every project can be housed with Arduino with all its shields. Some projects require too many communication protocols at once and need to perform a lot of I/O operations," says Hedge.

Fig. 1: Winkel Board



What Winkel Board does

Winkel shows up as a standard board under Arduino IDE. All you need to do is perform a standard board's manager installation. It also provides support to non-IDE users with avrdude programming, which runs on AVR command lines.

Unique and efficient. While other Arduino boards help you learn the basics, these do not offer what is really needed to get started with a real-world project prototype. Other boards also require you to find, add and configure other components yourself—a time-consuming, expensive and frustrating process.

Work together flawlessly. Winkel Board has a seamless working system. When you power on the board, the Wi-Fi module automatically goes into configuration mode, allowing you to connect the ESP module to your Wi-Fi network and retrieve an IP address assignment. Then, all you have to do is send a code to the module from any connected device.

A huge number of peripherals and modules on the board bring up the design challenge of bulkiness. The makers of Winkel Board have tried to pull off a two-sided PCB to fit everything in a form factor of 5cm × 5cm, thereby making the board compact and handy.